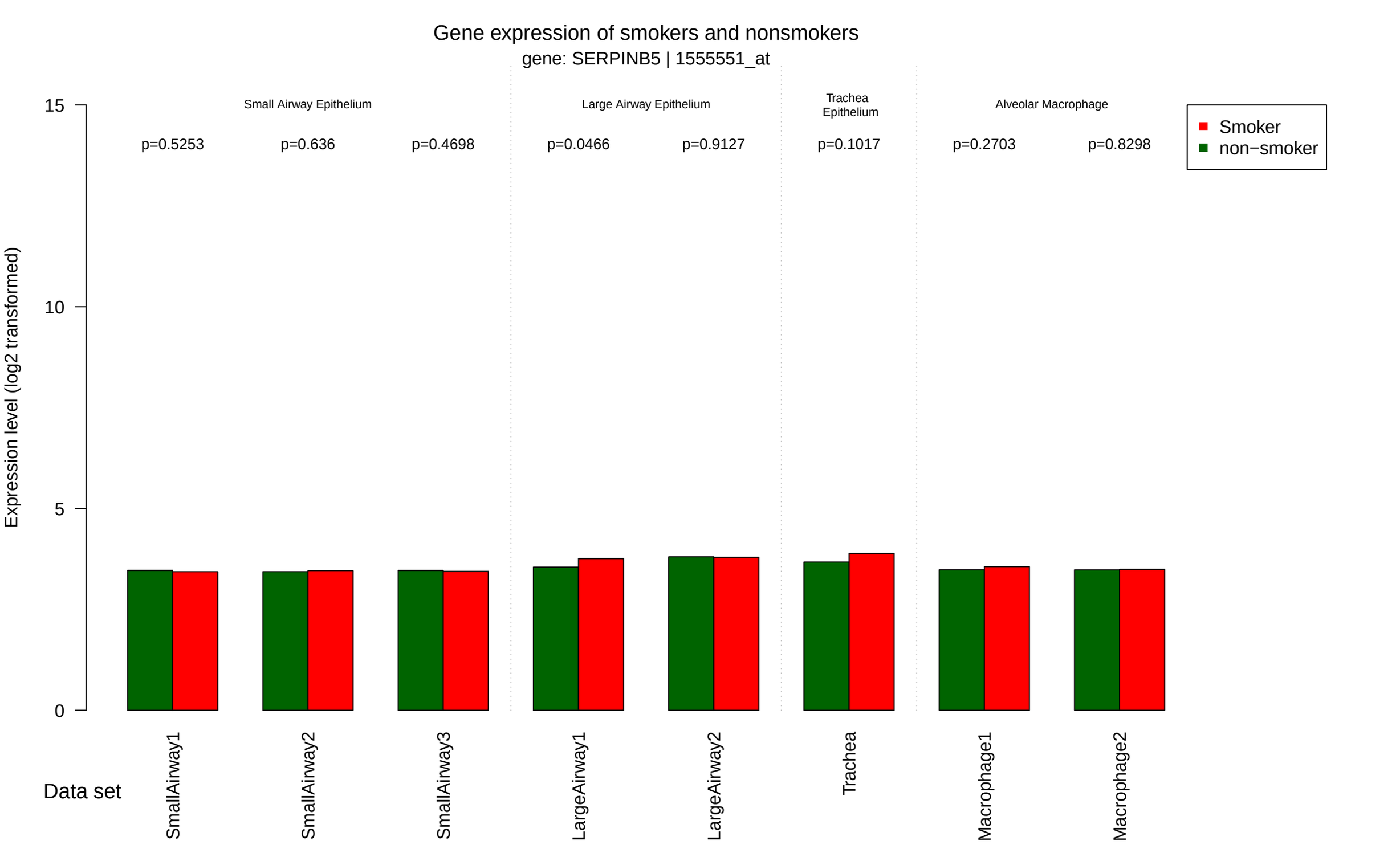
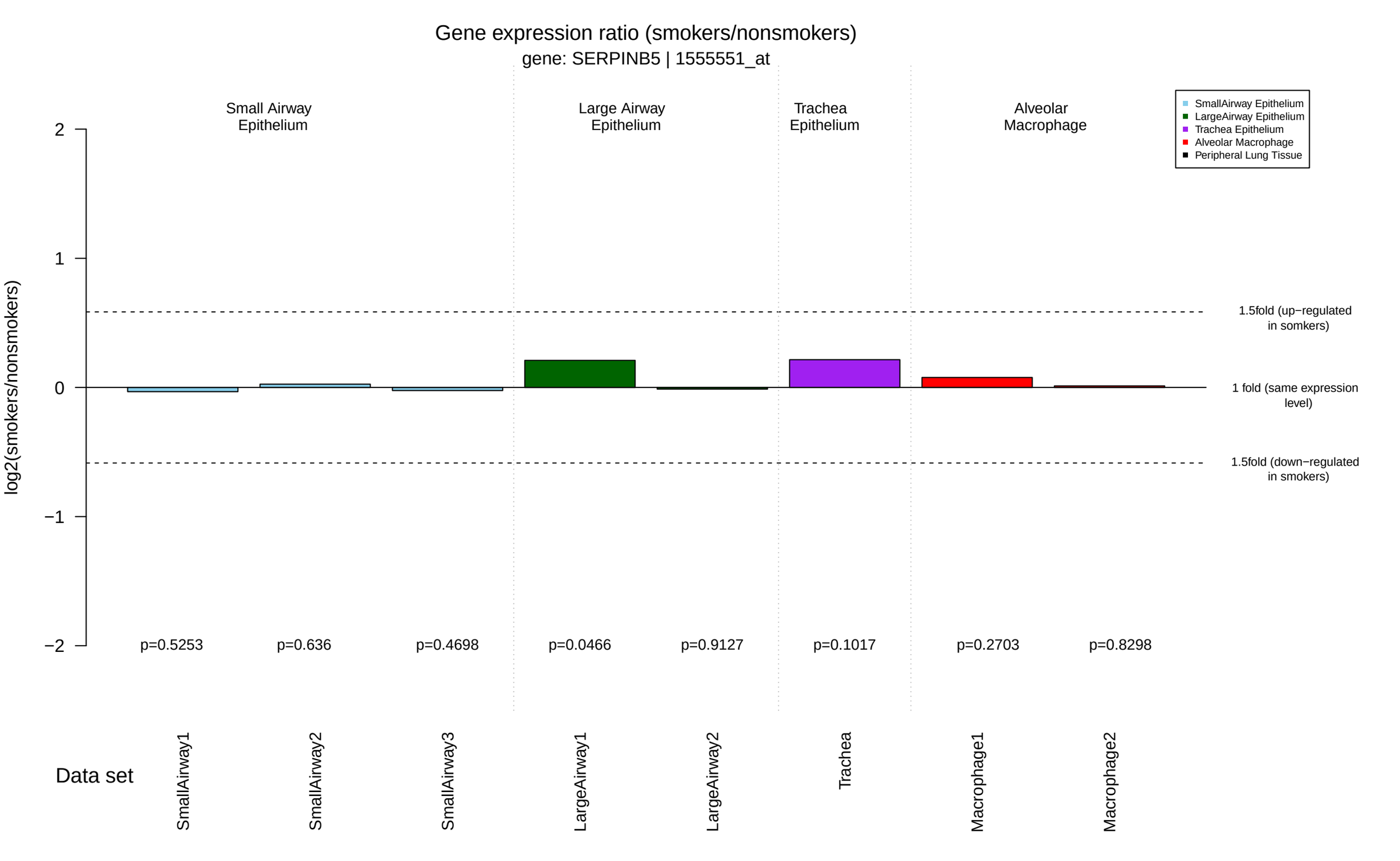
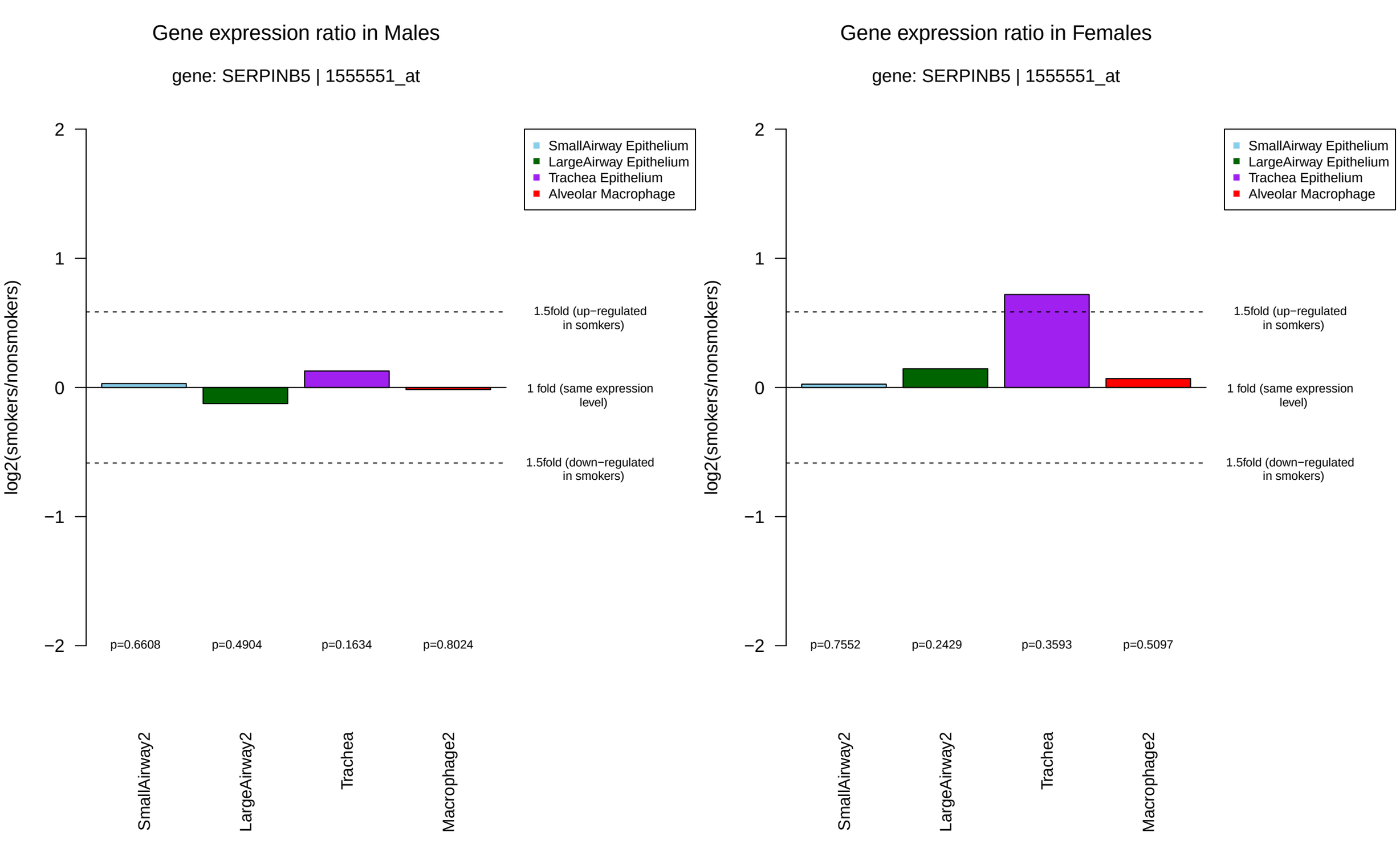
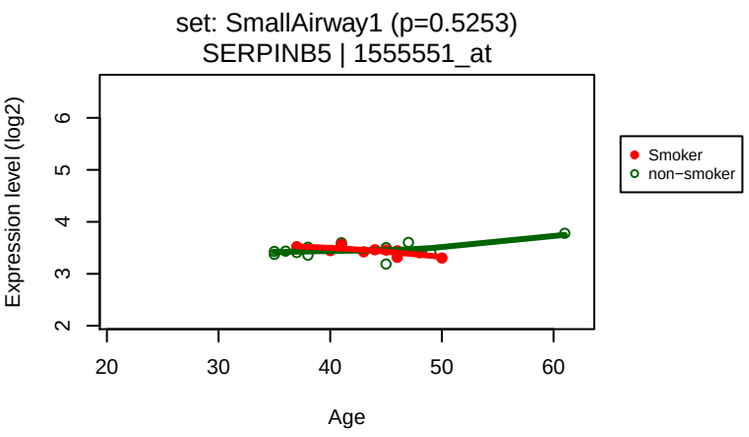
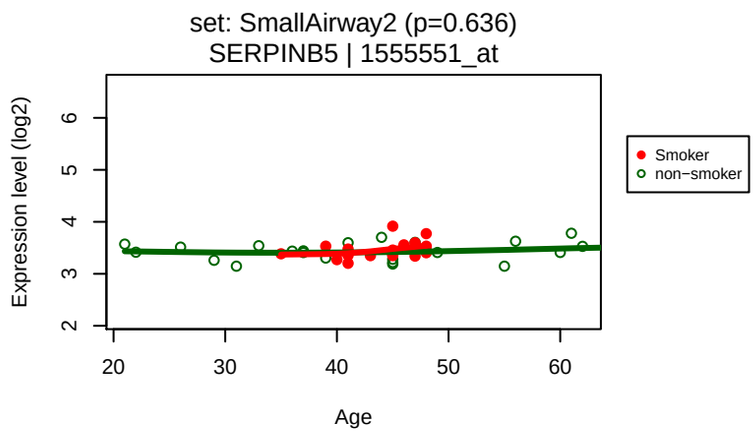
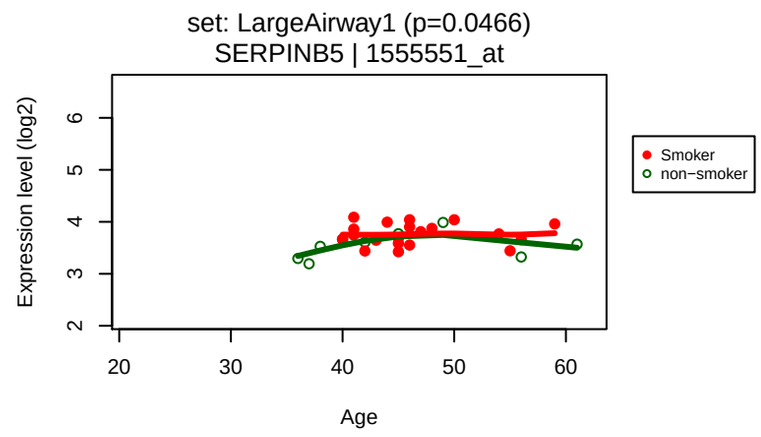

Gene expression of smokers and nonsmokers
gene: SERPINB5 | 1555551_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5253
p=0.636
p=0.4698
p=0.0466
p=0.9127
p=0.1017
p=0.2703
p=0.8298
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

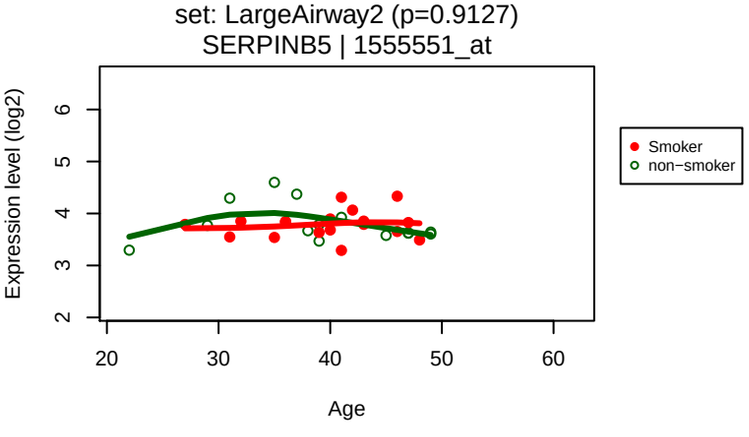

Gene expression ratio (smokers/nonsmokers)
gene: SERPINB5 | 1555551_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5253
p=0.636
p=0.4698
p=0.0466
p=0.9127
p=0.1017
p=0.2703
p=0.8298
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

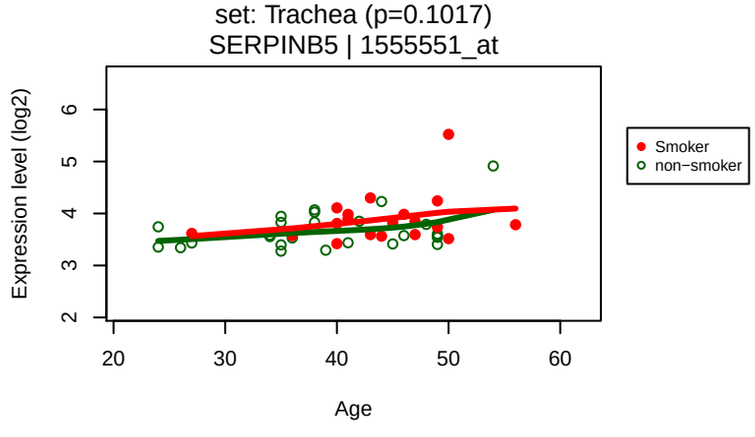

Gene expression ratio in Males
gene: SERPINB5 | 1555551_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6608
p=0.4904
p=0.1634
p=0.8024
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SERPINB5 | 1555551_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7552
p=0.2429
p=0.3593
p=0.5097
SmallAirway2
LargeAirway2
Trachea
Macrophage2

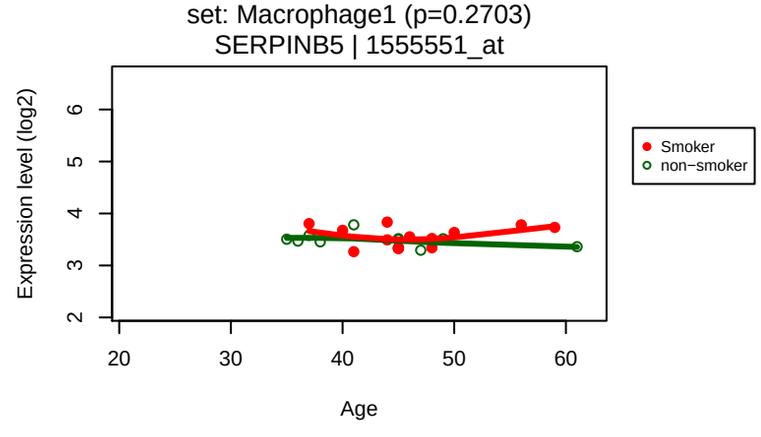

set: SmallAirway1 (p=0.5253)
SERPINB5 | 1555551_at
Expression level (log2)
Age
Smoker
non−smoker

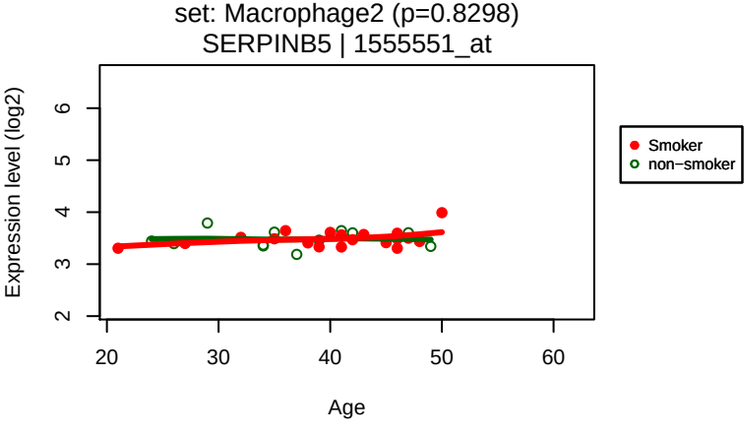

set: SmallAirway2 (p=0.636)
SERPINB5 | 1555551_at
Expression level (log2)
Age
Smoker
non−smoker

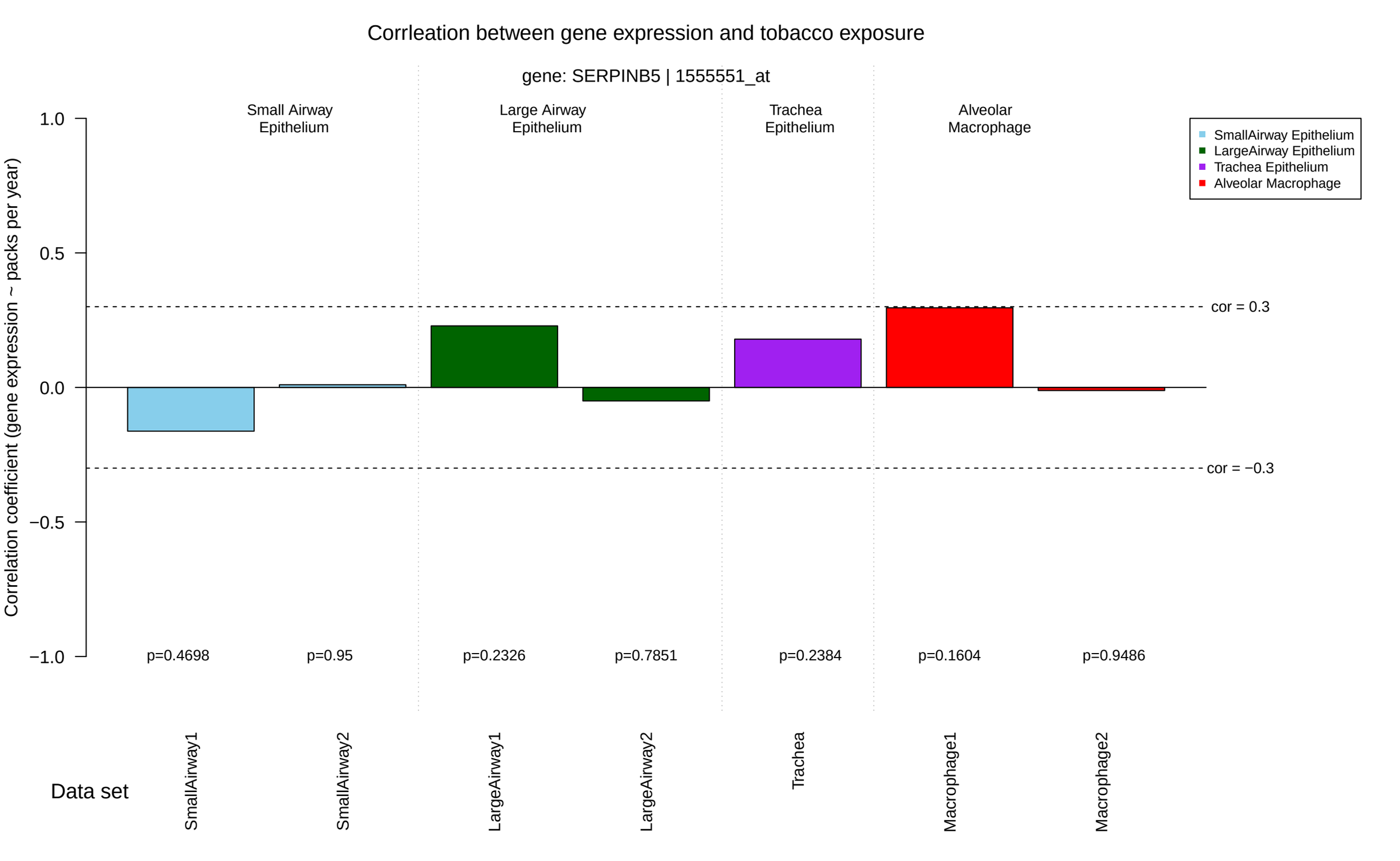

set: LargeAirway1 (p=0.0466)
SERPINB5 | 1555551_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9127)
SERPINB5 | 1555551_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1017)
SERPINB5 | 1555551_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2703)
SERPINB5 | 1555551_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8298)
SERPINB5 | 1555551_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SERPINB5 | 1555551_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4698
p=0.95
p=0.2326
p=0.7851
p=0.2384
p=0.1604
p=0.9486
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2